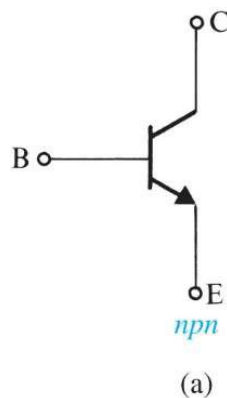


Outline

- › Physical structure
- › Current-voltage relationships
 - Regions (modes) of operation
- › BJT circuits

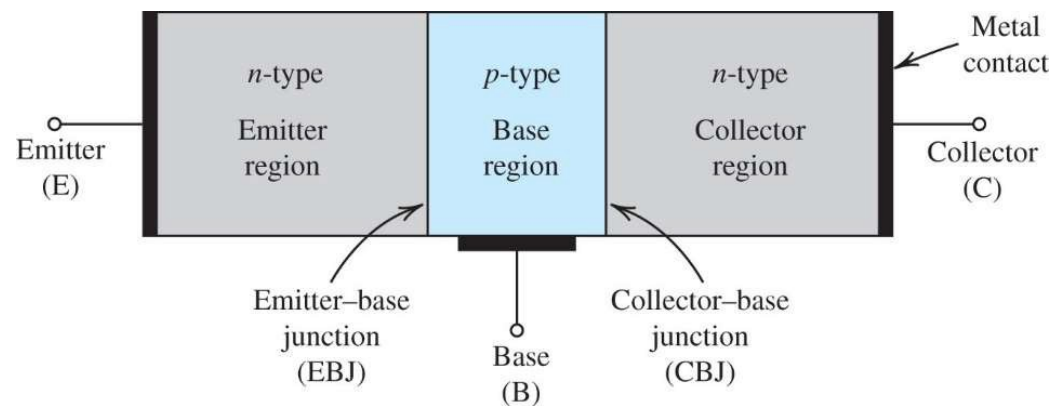
Introduction

- › BJT is a three-terminal device: **Base, Collector, Emitter**
- › Transistor (BJT) invented in 1948 at Bell labs
- › A new era of semiconductor devices and circuits
- › BJT not as common as MOSFETs in modern electronics
- › BJTs are used primarily in discrete-component (but also integrated) circuits



Device structure and physics

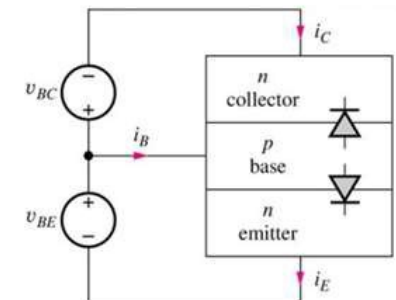
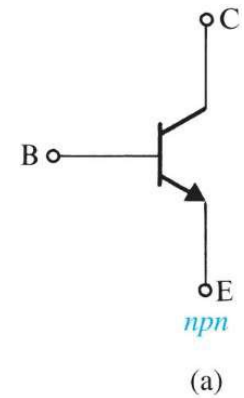
- › BJT consists of three semiconductor regions:
 - Emitter region: n-type
 - Base region: p-type
 - Collector region: n-type
- › Type described is npn. pnp types exist as well.



A simplified structure of the npn transistor

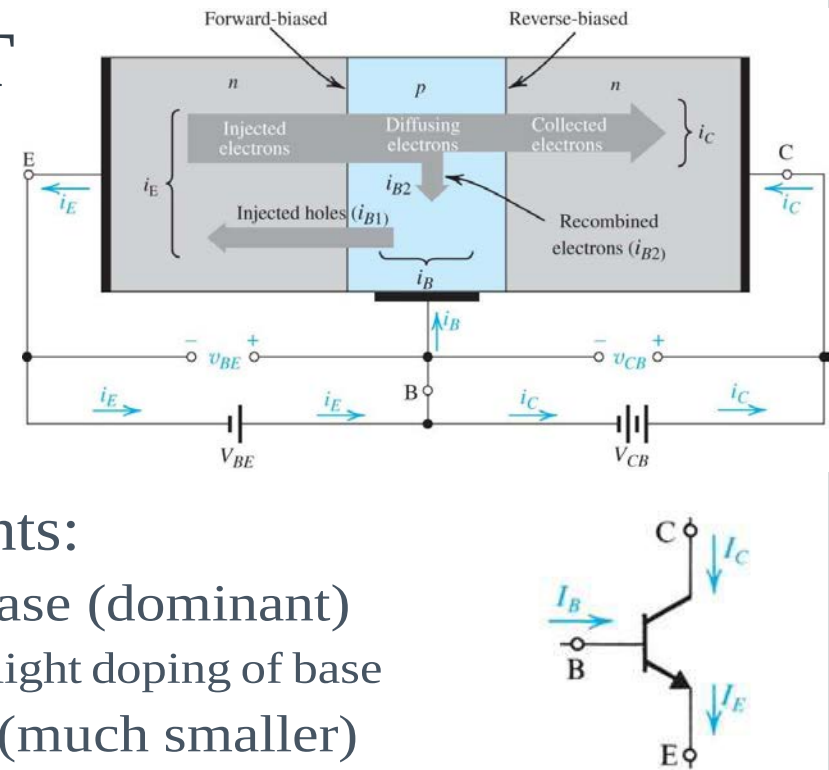
Modes of operation

- › Two pn junctions:
 - Base to emitter
 - Base to collector
- › Three modes of operation:
 - **Active** mode – used primarily for **amplification**
 - **Saturation** mode – used for primary for **switching**
 - **Cutoff** mode – device is off
- › Operating mode depends on biasing
- › Two external voltage sources are required for biasing
 - A source at the base (to bias B-E junction)
 - A source at the collector (to bias B-C junction)



Current flow in an npn BJT

- › Current flows when B-E junction is forward-biased
- › Recall that direction of current flow is opposite to the direction of electron flow
- › This current (i_E) has two components:
 - Electrons injected from emitter into base (dominant)
 - › is achieved with heavy doping of emitter, light doping of base
 - Holes injected from base into emitter (much smaller)
- › The electrons coming from the emitter will diffuse through the base and reach the collector
- › Thus, current flows out of the emitter and into the collector



Collector current in the forward active region

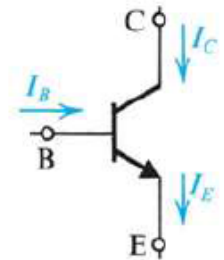
- › The current across a forward-biased pn junction is proportional to $e^{\frac{v_{BE}}{V_T}}$

- where v_{BE} is the forward voltage across the base-emitter junction
- and V_T = thermal voltage = $kT/q = 0.025$ V at room temperature

$$i_C = I_S e^{\frac{v_{BE}}{V_T}}$$

- where I_S = saturation current; typically 10^{-18} A $< I_S < 10^{-9}$ A. I_S is proportional to the area of the B-E junction and inversely proportional to the width, W , of the base region.

Note that i_C is independent of v_{BC} in the forward active region

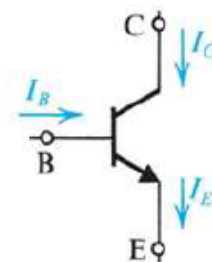


Base current in the forward active region

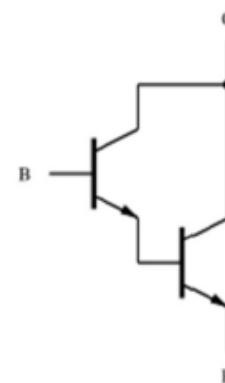
- › The base current in the **active region** can be expressed as:

$$i_B = \frac{i_C}{\beta}$$

$$i_B = \frac{I_S}{\beta} e^{\frac{v_{BE}}{V_T}}$$



- › β is known as current gain; typically, $20 < \beta < 500$
- › β can be higher for some special devices
- › A Darlington pair of BJTs increases β



Emitter current in the forward active region

- › Current that enters a transistor must leave it.

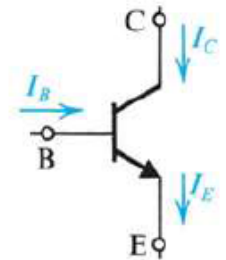
$$i_E = i_C + i_B$$

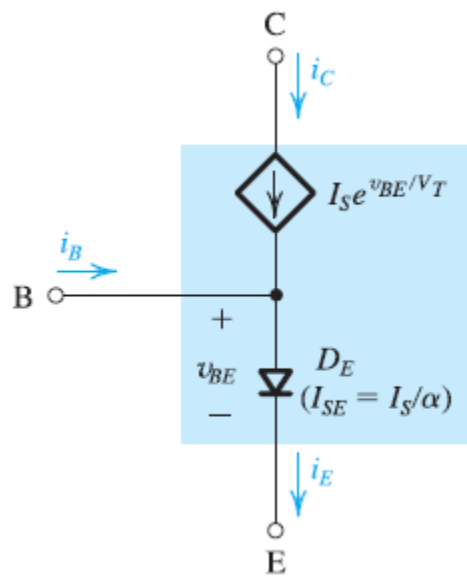
$$i_E = \frac{\beta + 1}{\beta} i_C = \frac{i_C}{\alpha}$$

$$\alpha = \frac{\beta}{\beta + 1} \text{ or } \beta = \frac{\alpha}{\alpha - 1}$$

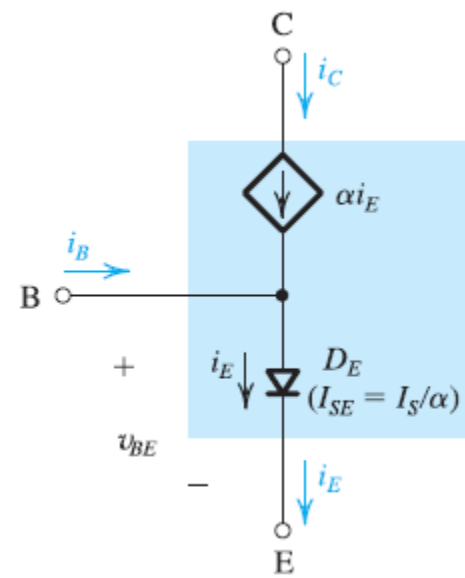
$$i_E = \frac{\beta + 1}{\beta} I_S e^{\frac{v_{BE}}{V_T}} = \frac{I_S}{\alpha} e^{\frac{v_{BE}}{V_T}}$$

- › parameter α : $0.95 < \alpha < 1$

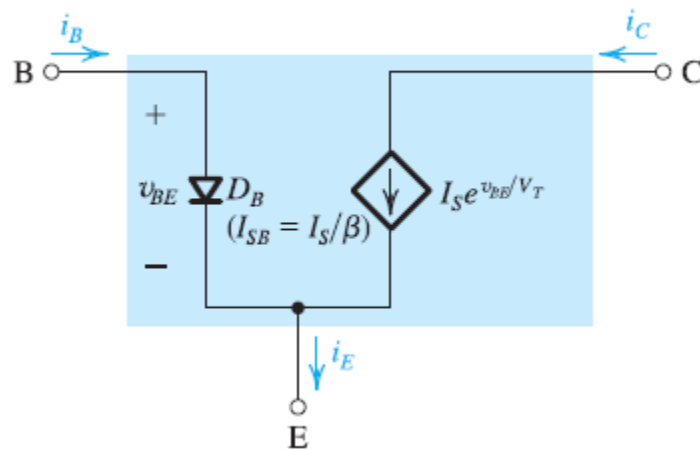




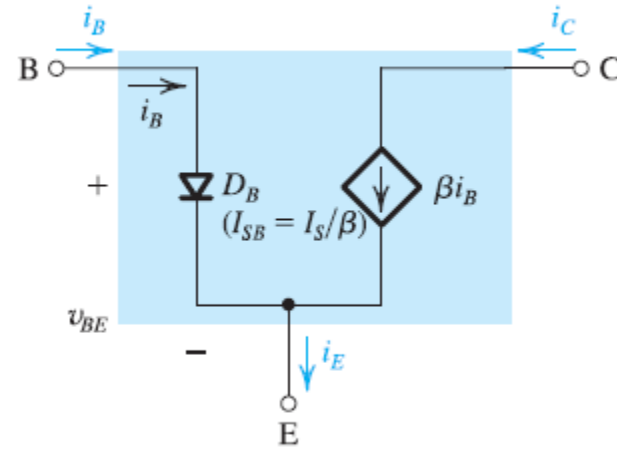
(a)



(b)



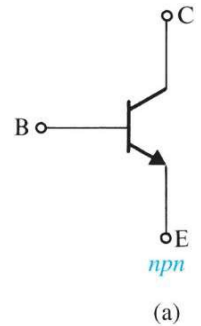
(c)



(d)

Large-signal equivalent-circuit models of the *n*pn BJT operating in the forward active mode.

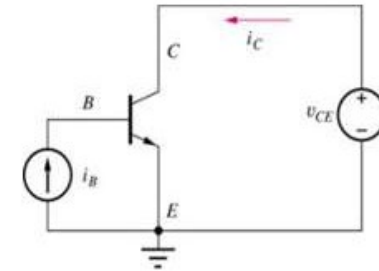
BJT (npn) modes of operation



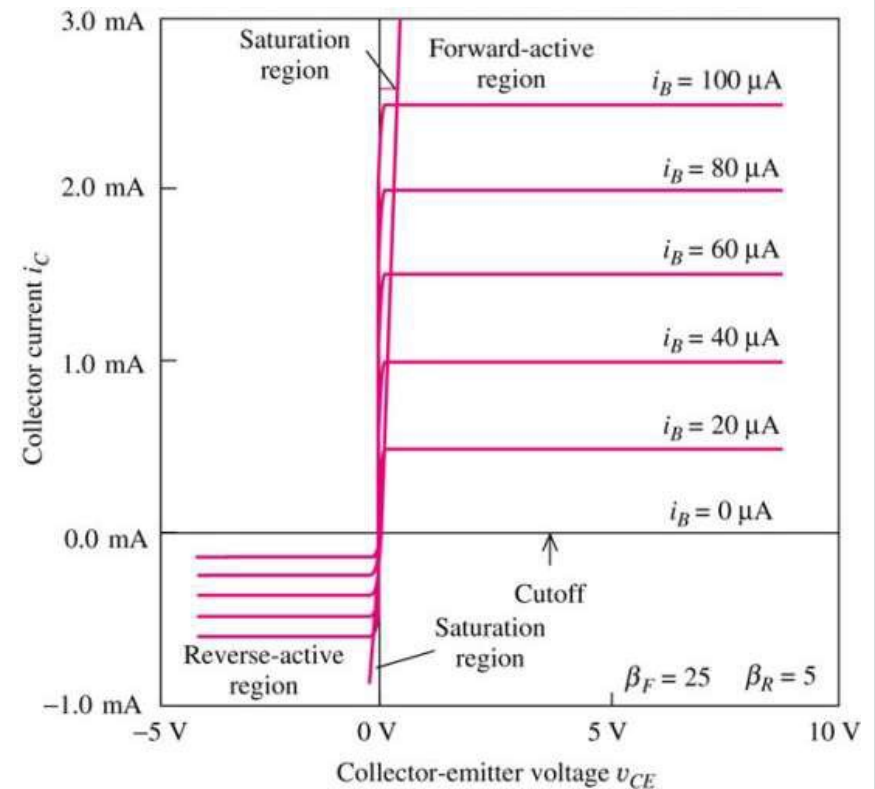
Base-Emitter Junction	Base-Collector Junction	
	Reverse Bias	Forward Bias
Forward Bias	Forward-Active Region (Good Amplifier)	Saturation Region (Closed Switch)
Reverse Bias	Cutoff Region (Open Switch)	Reverse-Active Region (Poor Amplifier)

IV characteristics of npn BJT

- › Common-emitter output
 - Apply a base current, vary V_{CE} and measure i_C
- › For $i_B = 0$, transistor is non-conducting or in **cutoff**
- › As i_B increases and for $V_{CE} > V_{BE}$ the transistor is in the **forward-active region**
 - i_C is almost independent of V_{CE}
- › For $V_{CE} < V_{BE}$, transistor is in **saturation** (narrow region)
 - V_{CE} is very small

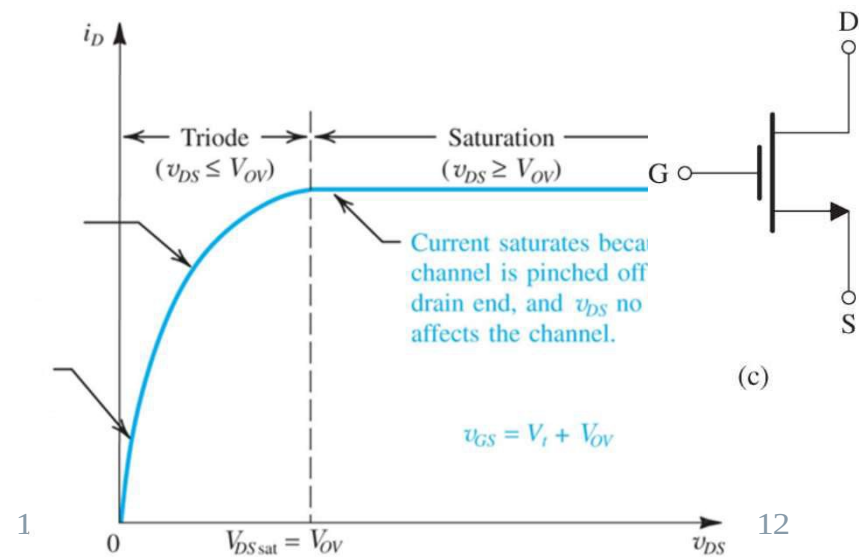
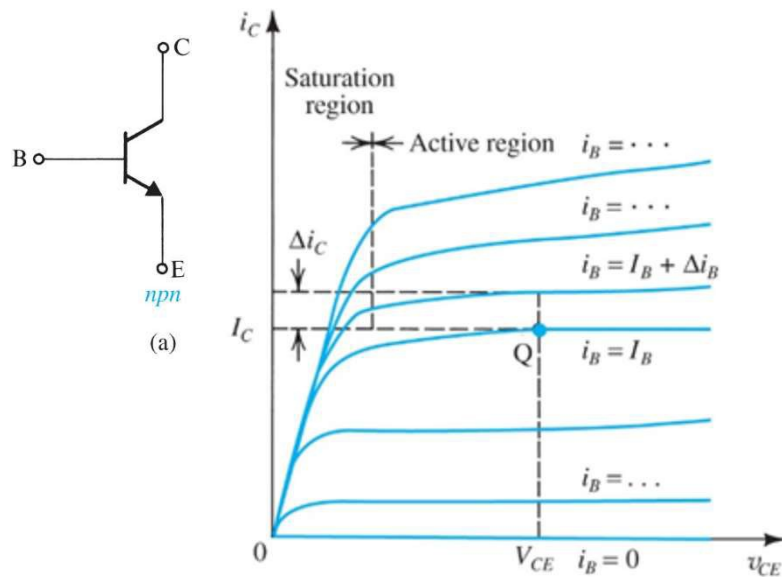


conceptual circuit for measuring i_C vs. V_{CE}



IV characteristics of npn BJT vs. NMOS

- Recall an NMOS with **gate**, **drain** and **source** and three operating regions: **saturation**, **triode** and **cutoff**
- Saturation mode of a BJT is completely different than saturation mode of a MOSFET
- Saturation mode of BJT corresponds to triode mode of MOSFET
- Active mode of BJT corresponds to saturation mode of MOSFET



Operational modes of the npn BJT

- › For BJT to operate in active mode, B-C junction must be reverse-biased
- › A pn junction does not become forward-biased until the forward voltage across it exceeds a set value ($\sim 0.4 \text{ V}$)
- › Active mode of operation of npn-BJT can be maintained down to $V_{BC} = \sim 0.4 \text{ V}$, or $V_{CB} = -0.4 \text{ V}$

Base-Emitter Junction	Base-Collector Junction	
	Reverse Bias	Forward Bias
Forward Bias	Forward-Active Region (Good Amplifier)	Saturation Region (Closed Switch)
Reverse Bias	Cutoff Region (Open Switch)	Reverse-Active Region (Poor Amplifier)

Operational modes boundary for the npn BJT

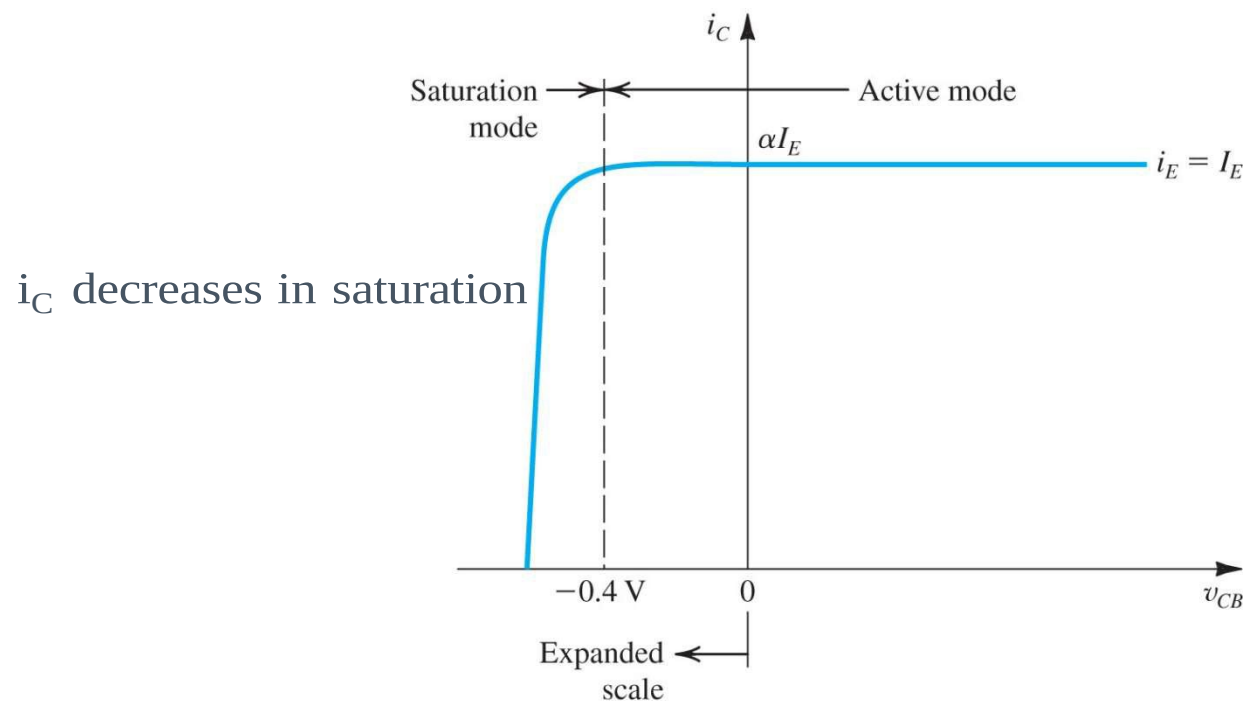
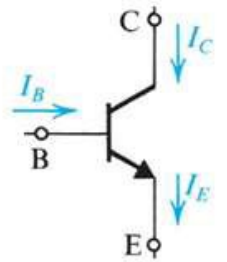


Figure 6.8 The $i_C - v_{CB}$ characteristic of an *npn* transistor fed with a constant emitter current I_E . The transistor enters the saturation mode of operation for $v_{CB} < -0.4$ V, and the collector current diminishes.

Operation in the saturation region

- › A BJT is a current-controlled device; the base current controls the BJT
- › When a small base current is applied, the BJT enters active region of operation
- › Recall that in active mode $i_C = \text{forward beta} = \beta_F i_B$
- › As the base current is increased further there comes a point where i_C does not equal $\beta_F i_B$; the BJT becomes 'saturated'
- › In saturation $i_C < \beta_F i_B$

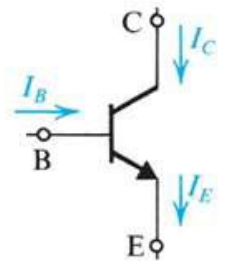


Operation in the saturation region

- › For a ‘saturated’ transistor the ratio i_C / i_B is lower than β_F
- › This ratio can be referred to ‘forced’ β

$$\beta_{forced} = \frac{i_C}{i_B} < \beta_F$$

- › For BJT to be in saturation:
 - B-C junction must be forward biased by more than 0.4 V

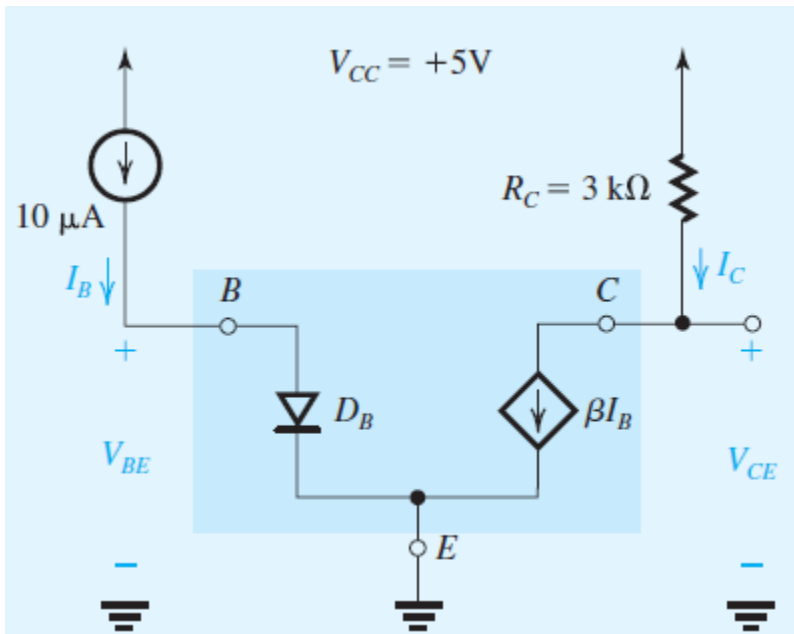


Base-Emitter Junction	Base-Collector Junction	
	Reverse Bias	Forward Bias
Forward Bias	Forward-Active Region (Good Amplifier)	Saturation Region (Closed Switch)
Reverse Bias	Cutoff Region (Open Switch)	Reverse-Active Region (Poor Amplifier)

Ex.

An NPN transistor having $I_s = 10^{-15} \text{ A}$ and $\beta = 100$ is connected as follows: The emitter is grounded, the base is fed with a constant-current source supplying a dc current of $10 \mu\text{A}$, and the collector is connected to a 5-V dc supply via a resistance R_c of $3 \text{ k}\Omega$.

Assuming that the transistor is operating in the active mode, find V_{BE} and V_{CE} .



$$\begin{aligned}
 V_{BE} &= V_T \ln \frac{I_B}{I_S / \beta} \\
 &= 25 \ln \left(\frac{10 \times 10^{-6}}{10^{-17}} \right) \\
 &= 690\ \text{mV} = 0.69\ \text{V}
 \end{aligned}$$

Next we determine the value of V_{CE} from

$$V_{CE} = V_{CC} - R_C I_C$$

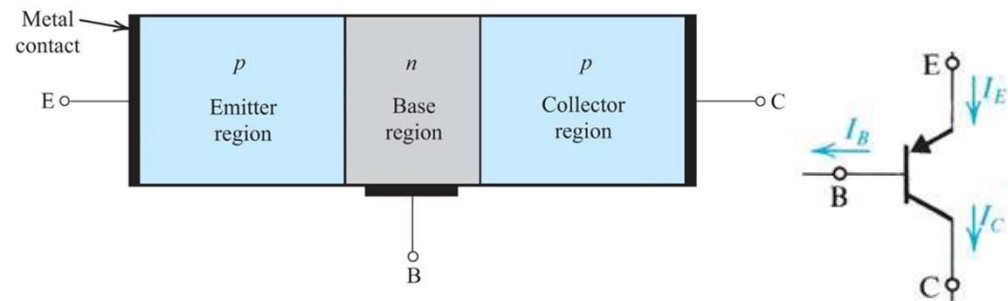
where

$$I_C = \beta I_B = 100 \times 10 \times 10^{-6} = 10^{-3}\ \text{A} = 1\ \text{mA}$$

Thus,

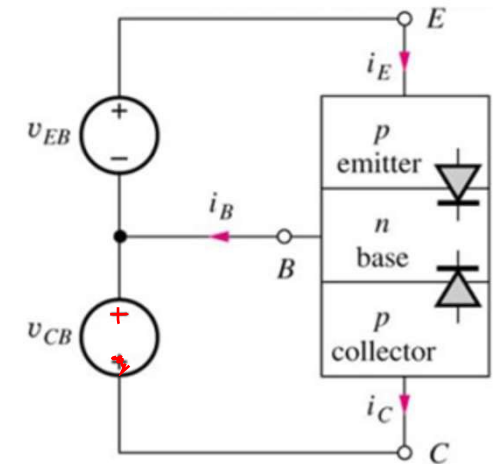
$$V_{CE} = 5 - 3 \times 1 = +2\ \text{V}$$

pnp transistor



- › p-type and n-type semiconductor material
- › Collector current and base current exit transistor terminals and emitter current enters the device
- › E-B junction is forward-biased and C-B junction is reverse-biased

Emitter-Base Junction	Collector-Base Junction	
	Reverse Bias	Forward Bias
Forward Bias	Forward-Active Region (Good Amplifier)	Saturation Region (Closed Switch)
Reverse Bias	Cutoff Region (Open Switch)	Reverse-Active Region (Poor Amplifier)



Summary of BJT characteristics

Table 6.2 Summary of the BJT Current–Voltage Relationships in the Active Mode

$$i_C = I_S e^{v_{BE}/V_T}$$

$$i_B = \frac{i_C}{\beta} = \left(\frac{I_S}{\beta} \right) e^{v_{BE}/V_T}$$

$$i_E = \frac{i_C}{\alpha} = \left(\frac{I_S}{\alpha} \right) e^{v_{BE}/V_T}$$

Note: For the *pnp* transistor, replace v_{BE} with v_{EB} .

$$i_C = \alpha i_E \qquad i_B = (1 - \alpha) i_E = \frac{i_E}{\beta + 1}$$

$$i_C = \beta i_B \qquad i_E = (\beta + 1) i_B$$

$$\beta = \frac{\alpha}{1 - \alpha} \qquad \alpha = \frac{\beta}{\beta + 1}$$

$$V_T = \text{thermal voltage} = \frac{kT}{q} \simeq 25 \text{ mV at room temperature}$$